

HACKATHON 7.0

Title of Hackathon: Develop a mobile based secure offline facial recognition and liveness detection system for remote locations.

The Objective: To develop a highly accurate, lightweight, and entirely offline facial recognition and liveness detection algorithm that can be seamlessly integrated into the existing Datalake 3.0 app, ensuring uninterrupted operations in zero-network zones.

The Problem Statement: "How can we accurately and securely authenticate field personnel using facial recognition and liveness detection on standard mid-range mobile devices without any active internet connection, while ensuring the AI model remains lightweight and seamlessly integrates with a React Native application on both android and iOS devices?"

Target Audience: Open to IT Students, AI/ML Enthusiasts, App Developers, Tech Professionals, and Startups

Technical Constraints & Specifications:

1. **Framework Compatibility:** Must be fully compatible with React Native, supporting cross-functional deployment on both Android and iOS.
2. **Model Footprint:** The AI model must be extremely lightweight to avoid bloating the Datalake app package. The target size is ~20 MB (the smaller, the better).
3. **Processing Speed:** Time taken to recognize a face and verify liveness must be < 1 second on standard mid-range devices.
4. **Hardware Requirements:** Must function smoothly without requiring high-end GPUs. Minimum supported OS: Android 8.0+ and iOS 12+ on devices with a minimum of 3GB RAM.
5. **Accuracy Threshold:** The facial recognition accuracy must be > 95%. The model must be trained to recognize diverse Indian demographics and function reliably in varying outdoor lighting conditions (e.g., harsh sunlight, low light, shadows).
6. **Open-Source Technologies only:** Solution should use only open-source technologies if at all any third-party code is used and source-code of the working prototype should be shared. No licenses should be required additionally.

Mandatory Deliverables

1. **Working Prototype with Source Code:** A functional cross-platform prototype (Android + iOS) built in React Native demonstrating the offline capability.
 - a. **Offline Liveness Detection:** The solution must include basic offline anti-spoofing measures (e.g., requiring the user to blink, smile, or turn their head slightly) to prevent attendance fraud via photographs or screens.
 - b. **Sync & Purge Mechanism:** The solution should have the scope for sync with AWS server after network connectivity is restored (local data to be purged)
2. **Presentation and Technical Documentation:** The solution should be well presented in a pptx. / .pdf file along with clear technical documentation detailing the model architecture, integration steps and performance benchmarks.

➤ **Submission Open Date: 22.05.2026**

➤ **Submission Closure Date: 05.06.2026**

Evaluation Criteria

1. **Innovation Level (30 Marks):** Efficiency of the edge AI model, compression techniques used to keep the size under 20MB, and effectiveness of offline liveness detection.
2. **Feasibility (30 Marks):** Ease of integration into the existing Datalake 3.0 React Native architecture and performance on mid-range devices (speed < 1 sec).
3. **Scalability & Sustainability (20 Marks):** Reliability of the offline-to-online sync/purge mechanism and adaptability to diverse lighting/demographics.
4. **Presentation & Documentation (20 Marks):** Clarity of the source code, integration guides, and final presentation to the evaluation committee.